

A Counterexample to Separable Determination of Fixed-Direction Sigma-Directional Porosity

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Source Question

The source paper is M. Cuth, M. Rmoutil and M. Zeleny, *On Separable Determination of Sigma-P-Porous Sets in Banach Spaces*, arXiv:1309.2174. On page 11, after Remark 4.8, the authors ask:

It is an open problem whether the notion of σ -directional porosity (see [12] for the definition) is separably determined in the sense of Corollary 5.11.

The same passage notes that the definition in Lindenstrauss–Preiss–Tiser is slightly different and equivalent to the cited definition only in separable Banach spaces. The counterexample below addresses the fixed-direction convention cited from Zajicek’s paper [12].

Remark 4.8. It is known to the authors that also the notions of lower porosity, $\langle g \rangle$ -porosity, and (g) -porosity satisfy the assumptions of Proposition 4.6 (for definitions see [9]). Consequently, those porosities (and corresponding σ -porosities) are separably determined when taking Suslin sets in complete metric spaces. We do not present proofs of those results here since now we see no interesting applications.

Note that in the following section we prove that also the notion of α -cone porosity in Asplund spaces satisfies the assumptions of Proposition 4.6 and, therefore, cone smallness is separably determined.

Question. *It is an open problem whether the notion of σ -directional porosity (see [12] for the definition) is separably determined in the sense of Corollary 5.11*

Note that the notion of σ -directional porosity is defined also in [5], but in a slightly different way which is equivalent to the definition from [12] only in separable Banach spaces.

5. CONE POROSITY

In the following section we prove that the notion of α -cone porosity in Asplund spaces satisfies the assumptions of Proposition 4.6 and, therefore, cone smallness is separably determined. First,

Definition Used

We use the fixed-direction convention indicated by the source paper. A set B in a Banach space X is porous in the direction $v \neq 0$ if there is $0 < c < 1$ such that, for every $b \in B$ and every $\varepsilon > 0$, there is a scalar t with $0 < \|tv\| < \varepsilon$ and

$$B_X(b + tv, c\|tv\|) \cap B = \emptyset.$$

A set is σ -directionally porous if it is a countable union of sets which are porous in fixed directions, with the direction allowed to vary from one member of the countable cover to another.

Proof Intuition

The obstruction is that a countable decomposition can use only countably many directions. In a nonseparable Hilbert space, countably many direction vectors miss some coordinate entirely. On the affine hyperplane where that coordinate equals one, all proposed directions are tangent. Therefore each proposed porous piece is nowhere dense inside that complete affine hyperplane, so Baire's theorem prevents the pieces from covering it.

Separable subspaces do not see this obstruction. A separable subspace of $\ell_2(\Gamma)$ is supported on countably many coordinates, and the example then becomes a countable union of proper affine hyperplanes, each porous in a fixed direction inside the subspace.

Theorem 1. *Let Γ be an uncountable set and let $X = \ell_2(\Gamma)$ over the real scalars. Define*

$$A = \{x \in X : x_\gamma = 1 \text{ for some } \gamma \in \Gamma\}.$$

Then A is closed, hence Suslin; A is not σ -directionally porous in X ; but for every separable linear subspace $V \subset X$, the set $A \cap V$ is σ -directionally porous in V .

Proof. First, A is closed. Suppose $x^m \in A$ and $x^m \rightarrow x$ in $\ell_2(\Gamma)$. For each m choose γ_m with $x_{\gamma_m}^m = 1$. If $\|x^m - x\| < 1/2$, then $|x_{\gamma_m}| > 1/2$. Since an $\ell_2(\Gamma)$ vector has only finitely many coordinates of modulus greater than $1/2$, one coordinate γ occurs along an infinite subsequence. Passing to that subsequence gives $x_\gamma = 1$, so $x \in A$.

We next prove that A is not σ -directionally porous. Assume, toward a contradiction, that

$$A = \bigcup_{n=1}^{\infty} A_n$$

where each A_n is porous in the fixed direction $v_n \neq 0$. Each $v_n \in \ell_2(\Gamma)$ has countable support, hence

$$S = \bigcup_{n=1}^{\infty} \text{supp } v_n$$

is countable. Choose $\gamma \in \Gamma \setminus S$ and consider the closed affine hyperplane

$$H_\gamma = \{x \in X : x_\gamma = 1\}.$$

Then $H_\gamma \subset A$, and v_n is tangent to H_γ for every n .

For each n , the set $A_n \cap H_\gamma$ is porous in the complete metric space H_γ , in the same direction v_n . Indeed, if $h \in A_n \cap H_\gamma$, then the holes for A_n centered at $h + tv_n$ have centers still in H_γ , because $(v_n)_\gamma = 0$; their intersections with H_γ are holes for $A_n \cap H_\gamma$. Such a porous subset is nowhere dense in H_γ : if some nonempty relatively open ball in H_γ were contained in its closure, taking a point of the set inside that ball and then a sufficiently small porosity hole would produce a smaller relatively open ball disjoint from the set, a contradiction.

Thus $H_\gamma = \bigcup_n (A_n \cap H_\gamma)$ is a countable union of nowhere dense subsets of the complete metric space H_γ , contradicting Baire's theorem. Therefore A is not σ -directionally porous in X .

It remains to check separable sections. Let $V \subset X$ be separable. Choose a countable dense subset $D \subset V$. The set

$$S_V = \bigcup_{d \in D} \text{supp } d$$

is countable, and $V \subset \ell_2(S_V)$. Hence

$$A \cap V = \bigcup_{\gamma \in S_V} \{x \in V : x_\gamma = 1\}.$$

For each $\gamma \in S_V$, the set $\{x \in V : x_\gamma = 1\}$ is either empty or a proper closed affine hyperplane of V . If it is nonempty, choose $w \in V$ with $w_\gamma \neq 0$. With $\varphi_\gamma(x) = x_\gamma$ on V , any point x in this affine hyperplane satisfies

$$\text{dist}_V(x + tw, \{y \in V : y_\gamma = 1\}) \geq \frac{|t| |\varphi_\gamma(w)|}{\|\varphi_\gamma\|}.$$

Thus the hyperplane is porous in the fixed direction w . Since S_V is countable, $A \cap V$ is σ -directionally porous in V . $\square$

Consequence

Taking $V_0 = \{0\}$, every closed separable subspace $V \supset V_0$ satisfies that $A \cap V$ is σ -directionally porous in V , while A is not σ -directionally porous in X . Hence the Corollary 5.11 style separable-determination statement fails for the fixed-direction σ -directional porosity convention cited in the source question.

Verification and Scope

The proof is purely analytic and uses no computation. The central checks are:

- A is closed, so it satisfies the source paper's Suslin hypothesis.
- Countably many proposed directions in $\ell_2(\Gamma)$ miss some coordinate γ .
- Baire's theorem applies to the complete affine Hilbert space H_γ .
- Every separable subspace of $\ell_2(\Gamma)$ is supported on countably many coordinates.

The result is stated for the fixed-direction convention referenced by the source paper through Zajicek [12]. It is not claimed to settle a different nonseparable convention in which the direction is allowed to vary point by point inside a single porous set.

Novelty Check

The run indexes were searched for arXiv:1309.2174 and for the phrases **sigma-directional porosity**, **directional porosity**, and **separably determined**; no existing packet or proof-gap entry for this question was found. The local arXiv source corpus was searched for the same phrases and found the source question plus related uses of σ -directionally porous sets, but no later local answer to this exact separable-determination problem. Web searches for exact phrases such as "**sigma-directional porosity**" "**separably determined**" and close variants did not return an explicit later solution. Human review should verify the definition in Zajicek's cited paper [12], because the source paper emphasizes that conventions differ outside separable spaces.

References

- [1] M. Cuth, M. Rmoutil and M. Zeleny, *On Separable Determination of Sigma-P-Porous Sets in Banach Spaces*, arXiv:1309.2174.
- [2] L. Zajíček, *On σ -porous sets in abstract spaces*, Abstract and Applied Analysis 2005, no. 5, 509–534.
- [3] J. Lindenstrauss, D. Preiss and J. Tiser, *Frechet differentiability of Lipschitz functions and porous sets in Banach spaces*, Annals of Mathematics Studies 179, Princeton University Press, 2012.